Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/27/2005
PI Name:	Hermanson, James C Ph.D.		
Project Title:	STABILITY AND FLOW STRUCTURE OF EVAPORATING FILMS IN REDUCED GRAVITY		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
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Space Biology Special Category:	None		
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Flight Program:			
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COI Name (Institution):	Allen, Jeffrey S. (Michigan Technological University) Bailey, Michael R (University of Washington)		
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	This experimental study of film evaporation involves three experimental configurations using identical test cells. The first fluid configuration is an evaporating film on a heated substrate. This configuration is expected to yield Marangoni and/or Rayleigh-Bénard convection in reduced gravity, with those behaviors modified by the action of evaporation. In the second configuration, heat will be provided directly to the liquid surface, for example by infrared heating. In this case evaporation is present but in the near absence of the forces that drive Marangoni and/or Rayleigh-Bénard convection. In the third configuration, a non-evaporating, heated film will be studied. In this case thermocapillary effects will potentially be present owing to the heating of the film, but effects of mass removal and vapor recoil due to vaporization will be removed. The film response is examined under both quasi-steady conditions and in the presence of a mechanical perturbation.		

Task Description:	Experimental diagnostics include detailed temperature and heat transfer measurements as well as ultrasonic measurement of film thickness. In addition, optical imaging is employed to determine the features of the resulting instabilities and associated changes in the cellular structure with within the film. The experiments will be supplemented by consideration of appropriate stability analyses of fluid layers undergoing evaporation. The objective of this research is to investigate the fundamental physics of film evaporation in reduced gravity. The evaporation rate, film stability, and interfacial motion are studied by isolating the effects due to surface tension gradients, buoyancy, latent heat, mass flux, and viscosity. This objective is being met using normal gravity experiments, reduced gravity experiments on NASA Reduced Gravity Aircraft, and through stability analysis. The evaporation of liquid on a heated surface is a phenomenon of importance in many engineering problems involving heat transfer and phase change. There has been relatively little research on the detailed mechanisms operative at a film interface undergoing evaporation in reduced-gravity conditions. Of particular interest and significance are the stability characteristics of the film, and how they differ from those of films of comparable scale in the absence of evaporation. Mechanisms that can impact the stability of evaporating films include thermocapillary mechanisms that drive instabilities and fluid motion in liquid layers with non-uniform temperature, as well as the vapor recoil associated with the mass loss due to evaporation. Both of these effects are expected to play a de-stabilizing role for film layers in reduced-gravity, where surface-tension-driven convection can become important. The high heat flux associated with phase change is an important consideration in spacecraft thermal control applications. Devices such as capillary pumped loops and heat pipes depend critically on evaporation and the successful utilization of these devices in space applications require a detailed understanding of evaporation phenomena under reduced gravity conditions. Other important technologies related to space exploration requiring a more in-depth understanding of evaporation include humidity control of cabin environment, water recovery from waste sources, oxygen production, water management in power systems such as fuel cells, and in-situ resource utilization for fuel production. Understanding evaporation phenomena is critical to the development and optimization of these space technologies which enable human exploration of space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The use of ultrasound for the dynamic monitoring of film thickness and growth rate developed as part of this effort has potentially widespread use for the determination of film behavior in a variety of two-phase flow devices. Research on evaporation mechanisms may assist in the optimization of many terrestrial technologies involving drying applications and coatings.
Task Progress:	Evaporation experiments have been conducted in a test cell containing a downwards-facing, cooled brass plate with an exposed diameter of 10 cm. The metal surface is polished to allow for double-pass shadowgraph imaging, and is instrumented with imbedded heat transfer gauges and thermocouples. A one-dimensional, finite-difference, inverse heat conduction program is employed to calculate the unsteady heat flux at the surface using data acquired by the imbedded heat transfer gauge. The model was validated by comparison with known temperatures and heat fluxes for the case of a dry surface. A pressure-control system controls the state of the saturated vapor. By varying the system pressure in a controlled fashion the film growth and heat transfer has been methodically examined over the full range of conditions, ranging from a thick, evaporating film exhibiting droplet break-off to a dry surface. A special test cell to rapidly initiate evaporation for Reduced Gravity Aircraft experiments has been developed and tested. Rapid heat transfer is achieved by several thermoelectric coolers and also by compression by a volume control system. The test cell is a 10 cm diameter, 31 cm long glass cylinder with an ultra-smooth copper plate at each end. The condensing test plate is 2.5 cm thick to maintain a constant wall temperature. The other plate, a 0.64 cm thick copper plate, is rapidly heated or cooled to provide vapor source/sink as needed to offset the system volume change. A feedback control system minimizes the system response time for testing in reduced-gravity conditions. Successful operation of the computer and feedback control systems has been demonstrated. An acoustic array measurement system is being employed for the measurement of the temporally- and spatially-varying evaporating film thickness. Recent effort has been directed at improving the spatial resolution of the thickness measurement by each individual element, and to balance bandwidth versus sensitivity to obtain a measurable signal of very short duration for optimal spatial resolution. The configuration developed to date includes a pulser/receiver generate an electrical spike as a drive signal, a transmit/receive switch and a broadband transducer. Acoustic signals have been generated and propagated through a 1-in (2.54-cm)-thick copper plate and reflecting at both interfaces of a stagnant, non-evaporating water film of known thickness resting on the top surface of the plate. A lip was milled on the edged of the plate to retain the water film. Film thicknesses smaller than 200 mm have been resolved to date, with measurement times on the order of 10 ms. The software-controlled measurement system is currently being modified to allow the use of multiple transducers in a transducer array.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Abstracts for Journals and Proceedings	Hermanson JC, Som SM, Allen JS, Pedersen PC. "Stability and Heat Transfer of Condensing and Evaporating Films." Strategic Research to Enable NASA’s Exploration Missions Conference and Workshop, Cleveland, OH, June 22-23, 2004. Strategic Research to Enable NASA’s Exploration Missions Conference and Workshop, Cleveland, OH, June 22-23, 2004. , Jun-2004
Dissertations and Theses	Som SM. "Stability and Heat Transfer Characteristics of an Unsteady Condensing Film." MS Thesis, University of Washington, December 2004. , Dec-2004
Papers from Meeting Proceedings	Som SM, Hermanson JC, Allen JS. "Stability and Heat Transfer of an Unsteady Condensing Film." 55th International Astronautical Congress, Vancouver, Canada, October 4-8, 2004. 55th International Astronautical Congress, Vancouver, Canada, October 4-8, 2004. Paper IAC-040-IAF-J.2.08. , Oct-2004